

Power and Bipower Variation with Stochastic Volatility and Jumps

OLE E. BARNDORFF-NIELSEN

University of Aarhus

NEIL SHEPHARD

University of Oxford, Nuffield College

ABSTRACT

This article shows that realized power variation and its extension, realized bipower variation, which we introduce here, are somewhat robust to rare jumps. We demonstrate that in special cases, realized bipower variation estimates integrated variance in stochastic volatility models, thus providing a model-free and consistent alternative to realized variance. Its robustness property means that if we have a stochastic volatility plus infrequent jumps process, then the difference between realized variance and realized bipower variation estimates the quadratic variation of the jump component. This seems to be the first method that can separate quadratic variation into its continuous and jump components. Various extensions are given, together with proofs of special cases of these results. Detailed mathematical results are reported in Barndorff-Nielsen and Shephard (2003a).

KEYWORDS: bipower variation, integrated variance, jump process, power variation, quadratic variation, realized variance, realized volatility, semimartingale, volatility

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The econometrics of financial volatility have made very significant recent progress due to the harnessing of high-frequency information through the use of realized variances. Here we discuss generalizations of objects of that type, studying sums of powers and products of powers of absolute returns. We will show that these quantities, called realized power variation and bipower variation, respectively, are both quite robust to rare jumps in the log-price process. In particular, we demonstrate that it is possible, in theory, to untangle the impact of the presence of volatility and rare jumps by using power and bipower variation. Realized bipower variation also provides a new asymptotically unbiased, model-free econometric estimator of integrated variance in stochastic volatility (SV) models. This estimator is robust to the presence of jumps. Hence, in theory, we can now decompose quadratic variation into the contribution from the continuous component of log-prices and the impact of jumps. Given that quadratic variation plays a crucial role in the theory of measuring the risk associated with an asset, being able to separate the diffusion and jump risks seems interesting. We believe ours is the first article to do this without making strong parametric assumptions.

To start, let us establish some notation. Suppose that $\bar{h} > 0$ is some fixed time period (e.g., a trading day or a calendar month) and that the log-price of an asset is written as $y^*(t)$ for $t \geq 0$. Then the i th $\bar{h}$ “low-frequency” return is

$$y_i = y^*(i\bar{h}) - y^*((i-1)\bar{h}), \quad i = 1, 2, \dots$$

For concreteness we will often refer to the i th period as the i th day. Suppose, in addition to y_i , that we record the prices at M equally spaced time points during the i th day. Then we can define the “high-frequency” returns as

$$y_{j,i} = y^*((i-1)\bar{h} + j\bar{h}M^{-1}) - y^*((i-1)\bar{h} + (j-1)\bar{h}M^{-1}), \quad j = 1, 2, \dots, M.$$

Here $y_{j,i}$ is the j th intra- $\bar{h}$ return for the i th day (e.g., if $M=288$, then this is the return for the j th five-minute period on the i th day). As a result, for example, $y_i = \sum_{j=1}^M y_{j,i}$.

To illustrate this notation we will look at the first three days of the Olsen and Associates’ dollar/DM high-frequency series. It starts on December 1, 1986, and ignores weekend breaks. This series is constructed every five minutes by the Olsen group from bid and ask quotes that appeared on the Reuters screen [see Dacorogna et al. (2001) for details]. The log values of the raw data, preprocessed in a manner discussed in the appendix of Barndorff-Nielsen and Shephard (2002b), are shown in the top left of Figure 1. We have shifted the data to start at zero at time zero. This series constitutes an empirical counterpart, recorded every five minutes, to our y^* process. The top right part of the figure shows the first three daily returns, $y_{1,i} = y_i$ (i.e., $M=1$), calculated off this series. Thus in this case $i=1, 2, 3$. These returns are shown as large squares. The bottom left part shows $y_{j,i}$ for $j=1, 2, \dots, 8$ and $i=1, 2, 3$. These are three-hour returns. Finally, the bottom right graph shows the effect of taking $M=48$, which corresponds to 30-minute returns.

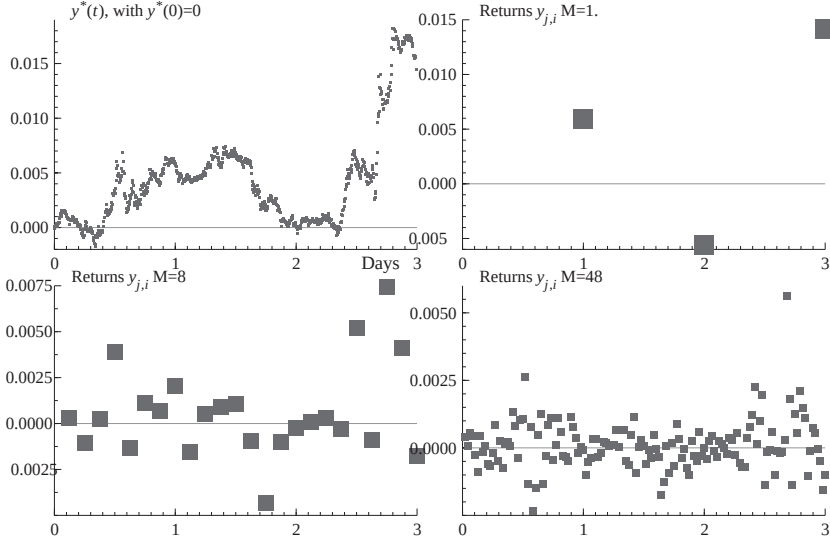


Figure 1 DM/dollar dataset from the Olsen and Associates database. Top left: Raw 5-minute data. Top right: Daily returns, $M=1$. Bottom left: 3-hour returns, $M=8$. Bottom right: 30-minute returns, $M=48$.

In this article we begin by recalling the well-known connections between *realized variance*,

$$\{y_M^*\}_i^{[2]} = [y_M^*]_i^{[2]} = \sum_{j=1}^M y_{j,i}^2, \quad (1)$$

and quadratic variation.¹ This is also linked to its square root version,² the *realized volatility*, $\sqrt{\sum_{j=1}^M y_{j,i}^2}$. The article then progresses to work with more general aggregate volatility measures. These will initially be based on *realized power variation*,

$$\{y_M^*\}_i^{[r]} = \left(\frac{\hbar}{M}\right)^{1-r/2} \sum_{j=1}^M |y_{j,i}|^r, \quad r > 0 \quad (2)$$

and, to a lesser extent, its unnormalized version,³ $[y_M^*]_i^{[r]} = \sum_{j=1}^M |y_{j,i}|^r$. These quantities have recently been formalized by Barndorff-Nielsen and Shephard (2003d).

¹ Realized variances have been used for a long time in financial economics [see e.g., Poterba and Summers (1986), Schwert (1989), and Dacoronga et al. (1998)]. Realized variance has been studied from a methodological viewpoint by, for example, Andersen et al. (2001) and Barndorff-Nielsen and Shephard (2001), (2002b). See Andersen, Bollerslev, and Diebold (2004) for a survey of this area, including a discussion of the related literature.

² Note that in econometrics sums of squared returns are sometimes called realized volatility.

³ The similarly named p -variation, $0 < p < \infty$, of a real-valued function f on $[a, b]$ is defined as

$$\sup_{\kappa} \sum |f(x_i) - f(x_{i-1})|^p,$$

where the supremum is taken over all subdivisions κ of $[a, b]$. If this function is finite, then f is said to have bounded (or finite) p -variation on $[a, b]$. The case of $p=1$ gives the usual definition of bounded variation. This concept of p -variation has been studied recently in the probability literature. See the work of Lyons (1994), Mikosch and Norvaiša (2000), and Lyons and Qian (2002).

A review of some of the work on this topic is given in Barndorff-Nielsen, Graversen, and Shephard (2003). When $r=2$, this yields the traditional *realized variance*. When $r=1$, we produce realized absolute variation,⁴ $\{y_M^*\}_i^{[1]} = \sqrt{(\hbar/M)} \sum_{j=1}^M |y_{j,i}|$.

The most important contribution of this article is the introduction of a generalization of the above ideas. This is called *realized bipower variation*,

$$\{y_M^*\}_i^{[r,s]} = \left\{ \left(\frac{\hbar}{M} \right)^{1-(r+s)/2} \sum_{j=1}^{M-1} |y_{j,i}|^r |y_{j+1,i}|^s, \quad r, s \geq 0. \right.$$

In particular, for example, we will show that in the $r=s=1$ case,

$$\{y_M^*\}_i^{[1,1]} = \sum_{j=1}^{M-1} |y_{j,i}| |y_{j+1,i}| \quad (3)$$

will, up to a simple known multiple, converge to the same probability limit as realized variance when prices follow a SV process and that for Equation (3) the limit does not change with the addition of rare jumps. This provides a new way of making inferences and predicting integrated variance in SV models, perhaps the single most important term in the econometrics of volatility. Further, it gives a simple way of measuring the impact of jumps on quadratic variation.

In Section 1 we review the well-known probability limit of realized variance, which is built on the theory of semimartingales and the quadratic variation process. We specialize the theory to a class of continuous sample path stochastic volatility semimartingales. For this class, it is possible to extend the quadratic variation process to the power variation process, which allows us to derive the probability limit of realized power variation. Finally, we introduce the idea of bipower variation and study some of its properties.

Having established the probability limit for the various realized variations, in Section 2 we find the probability limit of realized power variation in the case where we add a compound Poisson process to the SV process. We see that sometimes the limit is not changed by the addition of jumps. We then extend this analysis to realized bipower variation and see that the robustness result holds so long as the maximum of the powers is less than two. This result is the main contribution of the article.

In Section 3 we conduct a simulation study of a SV plus jump process, demonstrating that the theory seems to yield useful predictions. In Section 4 we apply the theory to some empirical data and in Section 5 we indicate various possible extensions of our work. Section 6 concludes.

⁴ Andersen and Bollerslev (1997, 1998b) empirically studied the properties of sums of absolute values of intraday returns on speculative assets. This was empirically attractive, since using absolute values is less sensitive to possible large movements in high-frequency data. However, the approach was abandoned in the subsequent work reported in Andersen and Bollerslev (1998a) and Andersen et al. (2001) due to the lack of appropriate theory for the sum of absolute returns as $M \rightarrow \infty$, although recently Andreou and Ghysels (2002) have performed some interesting Monte Carlo studies in this context, while Maheswaran and Sims (1993) and Shiryayev (1999, pp. 349–350) mention interest in the limit of sums of absolute returns.

1 BASIC DEVELOPMENT AND IDEAS

1.1 Realized Variance and Quadratic Variation

The i th realized variance is based on squares of high-frequency observations in the interval of time between $h(i-1)$ and hi . To understand the properties of realized variance we need to connect it to the quadratic variation process — which we will define in a moment. It is traditional, convenient, and elegant to write the theory of quadratic variation based on the interval from time 0 to t . We will follow that route here and continue with it when we initially discuss its generalizations to power variation. We can then back out the implication for items such as realized variance, which are based solely on quantities calculated on the interval from $h(i-1)$ to hi . These specific realized variance results will be presented at the end of this subsection.

The probability limit of realized variance in Equation (1) is known under the assumption that y^* is a semimartingale (SM), using the theory of quadratic variation [e.g., Jacod and Shiryaev (1987, p. 55)]. Recall if $y^* \in SM$, then we can write

$$y^* = \alpha^* + m^*, \quad (4)$$

where α^* , a drift term, has *locally finite variation* paths ($\mathcal{FV}_{loc}$) and m^* is a *local martingale* ($\mathcal{M}_{loc}$). One of the most important aspects of semimartingales is the *quadratic variation* (QV) process. This is defined as

$$[y^*](t) = p - \lim_{M \rightarrow \infty} \sum_{j=1}^M \{y^*(t_j) - y^*(t_{j-1})\}^2, \quad (5)$$

for any sequence of partitions $t_0 = 0 < t_1 < \dots < t_M = t$ so long as $\sup_j \{t_j - t_{j-1}\} \rightarrow 0$ for $M \rightarrow \infty$.

In general,

$$\begin{aligned} [y^*](t) &= [y^{*c}](t) + [y^{*d}](t) \\ &= [y^{*c}](t) + \sum_{0 \leq s \leq t} \{\Delta y^*(s)\}^2, \end{aligned}$$

where y^{*c} is the continuous local martingale component of y^* , y^{*d} is the corresponding jump process, and $\Delta y^*(t) = y^*(t) - y^*(t-)$ is the jump at time t . If $\alpha^* \in \mathcal{FV}_{loc}^c$ (i.e., the subset of the $\mathcal{FV}_{loc}$ class that has continuous sample paths), then we obtain the simplification

$$[y^*](t) = [m^*](t) = [m^{*c}](t) + \sum_{0 \leq s \leq t} \{\Delta m^*(s)\}^2.$$

It is somewhat instructive to think of the economics of semimartingales and quadratic variation in the following way. We start with $y^* \in SM$ so that

$$dy^* = d\alpha^* + dm^*.$$

If $\alpha^* \in \mathcal{FV}_{loc}^c$ (and so predictable) and if $m^* \in \mathcal{M}$ (a martingale), then the predictability of α^* means that

$$\mathbb{E}(dy^*(t)|\mathcal{F}(t)) = d\alpha^*(t),$$

where $\mathcal{F}(t)$ is the natural filtration of $y^*(t)$. Thus the drift can be thought of as

$$\alpha^*(t) = \int_0^t \mathbb{E}(dy^*(u)|\mathcal{F}(u)),$$

the integral of the expected instantaneous returns. In addition, so long as $m^* \in \mathcal{M}^c$ is locally square integrable, then

$$\begin{aligned} [y^*](t) &= \int_0^t \text{Var}(dy^*(u)|\mathcal{F}(u)) \\ &= \int_0^t \text{Var}(dm^*(u)|\mathcal{F}(u)), \end{aligned}$$

that is, quadratic variation equals the integrated variance of the instantaneous returns. More generally, allowing jumps in the martingale component of prices, so long as $\alpha^* \in \mathcal{FV}_{loc}^c$, we have that

$$[y^*](t) = \int_0^t \text{Var}(dm^{*c}(u)|\mathcal{F}(u)) + \sum_{0 \leq s \leq t} \{\Delta m^*(s)\}^2.$$

Andersen et al. (2003) discuss the connection between $[y^*](t)$ and $\text{Var}(m^*(t)|\mathcal{F}(0))$, which is important from a practical forecasting viewpoint.

The above analysis implies that if $y^* \in \mathcal{SM}$,

$$[y_M^*]_i^{[2]} = \sum_{j=1}^M y_{j,i}^2 \xrightarrow{p} [y^*](hi) - [y^*](h(i-1)) = [y^*]_i,$$

meaning that realized variance consistently estimates daily increments of QV. In turn, so long as $\alpha^* \in \mathcal{FV}_{loc}^c$, then

$$[y_M^*]_i^{[2]} \xrightarrow{p} [m^*]_i.$$

In addition, if $\alpha^* \in \mathcal{FV}_{loc}^c$ and m^* is square integrable, then

$$[y_M^*]_i^{[2]} \xrightarrow{p} [m^{*c}]_i + [m^{*d}]_i = \int_{h(i-1)}^{hi} \text{Var}(dm^{*c}(u)|\mathcal{F}(u)) + [m^{*d}]_i,$$

the integrated variance of the instantaneous returns of the continuous component plus the quadratic variation of the jump part of m^* . This implies, importantly, that realized variance cannot be informative about the separate contributions the diffusion or jump parts of m^* make to QV. Instead, in order to break up these two components, we either have to make very specific parametric assumptions about y^* [see, e.g., Eraker, Johannes, and Polson (2003)] or use alternative aggregate volatility statistics. We follow the latter route in this article.

1.2 Stochastic Volatility

To extend the results on quadratic variation we will need some additional smoothness assumptions. We start by stating two, after recalling that $y^* = \alpha^* + m^*$:

Assumption 1 m^* is a SV process,⁵

$$m^*(t) = \int_0^t \sigma(u) dw(u), \quad (6)$$

where w is standard Brownian motion, $\sigma(t) > 0$, the spot volatility process, is càdlàg and locally bounded away from zero, and the integrated variance process,

$$\sigma^{2*}(t) = \int_0^t \sigma^2(u) du, \quad (7)$$

satisfies $\sigma^{2*}(t) < \infty$ for all $t < \infty$.

Assumption 2 The mean process α^* is continuous.

We call semimartingales satisfying Assumptions 1 and 2 members of the *continuous SV semimartingales* class (denoted $\mathcal{SVSM}^c$). Clearly $\mathcal{SVSM}^c \subset \mathcal{SM}^c \subset \mathcal{SM}$, where $\mathcal{SM}^c$ is the class of continuous semimartingales.⁶

Of importance is that, if $y^* \in \mathcal{SVSM}^c$, then

$$[y^*](t) = \sigma^{2*}(t). \quad (8)$$

Under the additional assumption that σ is continuous,⁷ $\partial[y^*](t)/\partial t = \sigma^2(t)$. Finally, for all $y^* \in \mathcal{SVSM}^c$,

$$\{y_M^*\}_i^{[2]} = [y_M^*]_i^{[2]} = \sum_{j=1}^M y_{j,i}^2 \xrightarrow{p} \sigma_i^2, \quad \text{where } \sigma_i^2 = \sigma^{2*}(\bar{h}i) - \sigma^{2*}(\bar{h}(i-1)).$$

We call σ_i^2 and $\sqrt{\sigma_i^2}$ the *actual variance* and *actual volatility*, respectively, of the SV component over the i th interval. All these results on SV processes are very well known. They hold irrespective of the relationship between α^* , σ , and w .

⁵ The literature of SV models is discussed by, for example, Harvey, Ruiz, and Shephard (1994), Taylor (1994), Ghysels, Harvey, and Renault (1996), Shephard (1996), and Kim, Shephard, and Chib (1998).

⁶ This construction looks, at first sight, ad hoc. However, this is not true. All local martingales with absolutely continuous quadratic variation can be written in the form of assumption (1). This result, which is from Doob (1953), is discussed in, for example, Karatzas and Shreve (1991, pp. 170–172). If $m^* \in \mathcal{M}_{loc}^c$ and α^* is predictable, then, if we impose a lack of arbitrage opportunity, $\alpha^* \in \mathcal{FV}_{loc}^c$. This condition is discussed in Andersen et al. (2003, p. 583). Hence, in the context of financial economics, one might argue that Assumption 2 follows from Assumption 1. From the Dambis-Dubins-Schwartz theorem we know that the difference between $\mathcal{M}_{loc}^c$ and those processes given in Assumption 1 are the local martingales that have only continuous, not absolutely continuous, quadratic variation. Hence Assumptions 1 and 2 are only slightly stronger than saying prices must have continuous sample paths.

⁷ More generally, if $D_-g(x)$ denotes the left derivative of a function g , $\lim_{\varepsilon \downarrow 0} \varepsilon^{-1} \{g(x) - g(x - \varepsilon)\}$, then $D_-[y^*](t) = \sigma^2(t-)$. This is relevant when the σ process exhibits jumps [e.g., Barndorff-Nielsen and Shephard (2001) and Eraker, Johannes, and Polson (2003)].

1.3 Power Variation Process

The quadratic variation process $[y^*]$ was generalized to the *power variation* process by Barndorff-Nielsen and Shephard (2003d). Unlike the QV case, power variation is, at least initially [see Barndorff-Nielsen and Shephard (2003c) and Barndorff-Nielsen, Graversen, and Shephard (2004) for a discussion of the more general case], defined via an equally spaced discretization of time. Again we work on the time interval from 0 to t , but now we assume we have observations every $\delta > 0$ periods of time. Our interest will be in seeing what happens as $\delta \downarrow 0$. This will determine the probability limit of realized power variation $\{y_M^*\}_t^{[r]}$ defined in Equation (2), which will be given explicitly at the end of this subsection.

The r th order power variation process ($r > 0$) is defined, when it exists, as

$$\{y^*\}_t^{[r]} = p - \lim_{\delta \downarrow 0} \delta^{1-r/2} \sum_{j=1}^{\lfloor t/\delta \rfloor} |y_j(t)|^r,$$

where the equally spaced j th return is

$$y_j = y_j(t) = y^*(j\delta) - y^*((j-1)\delta).$$

Here, for any real number a , $\lfloor a \rfloor$ denotes the largest integer less than or equal to a . The normalization $\delta^{1-r/2}$ is essential in power variation.

1. When $r = 2$ the normalization is one and so disappears.
2. When $r > 2$ the normalization goes off to infinity as $\delta \downarrow 0$.
3. When $r < 2$ the normalization goes to zero as $\delta \downarrow 0$.

The key property of power variation for $SVSM^c$ processes is given as follows.

Theorem 1 *If $y^* \in SVSM^c$ and additionally (σ, α^*) are independent of w , then*

$$\{y^*\}_t^{[r]} = \mu_r \int_0^t \sigma^r(s) ds, \quad \text{where } \mu_r = E|u|^r = 2^{r/2} \frac{\Gamma(\frac{1}{2}(r+1))}{\Gamma(\frac{1}{2})},$$

for $r > 0$ where $u \sim N(0, 1)$.

Proof See Barndorff-Nielsen and Shephard (2003d). ■

Remark 1 It is thus far unclear to us how substantial is the impact of relaxing the independence condition on (σ, α^*) and w . Note that no additional assumptions are made on the (σ, α^*) process beyond those stated in Assumptions 1 and 2.

The theorem implies that if σ is continuous,⁸ $(\mu_r^{-1} \partial \{y^*\}_t^{[r]} / \partial t)^{1/r} = \sigma(t)$.

The probability limit of realized power variation follows immediately from the definition of the power variation process. Thus, under the conditions of

⁸ Otherwise $(\mu_r^{-1} D_- \{y^*\}_t^{[r]}(t))^{1/r}$ is $\sigma(t-)$.

Theorem 1, as $M \rightarrow \infty$,

$$\begin{aligned} \{y_M^*\}_i^{[r]} &\xrightarrow{p} \{y^*\}^{[r]}(\bar{h}i) - \{y^*\}^{[r]}(\bar{h}(i-1)) \\ &= \mu_r \int_{\bar{h}(i-1)}^{\bar{h}i} \sigma^r(s) ds. \end{aligned} \quad (9)$$

Barndorff-Nielsen and Shephard (2003d) have extended the above convergence in probability to a distribution theory for $\{y^*\}^{[r]}(t) - \mu_r \int_0^t \sigma^r(s) ds$. This will be discussed in Section 5.4 of this article. Andreou and Ghysels (2003) have used power variations to test for changes in the level of volatility in financial markets.

1.4 Bipower Variation Process

In the context of multivariate covariation, Barndorff-Nielsen and Shephard (2002a) sometimes found it helpful to study cross terms of the type

$$\delta^{-1} \sum_{j=1}^{\lfloor t/\delta \rfloor - 1} y_j^2 y_{j+1}^2 \xrightarrow{p} \int_0^t \sigma^4(u) du,$$

where we recall that $y_j = y^*(j\delta) - y^*((j-1)\delta)$. Here we provide an extension of this that will allow us to understand realized bipower variation, the key innovation in this article. First we define the *bipower variation* (BPV) process as

$$\{y^*\}^{[r,s]}(t) = p - \lim_{\delta \downarrow 0} \delta^{1-(r+s)/2} \sum_{j=1}^{\lfloor t/\delta \rfloor - 1} |y_j|^r |y_{j+1}|^s, \quad r, s \geq 0,$$

provided this exists. We will define a more general k th order power variation process in Section 5.2. Clearly the value of $r+s$ is crucial here, as it determines the δ normalization. Importantly,

$$\{y^*\}^{[r,0]}(t) = \{y^*\}^{[0,r]}(t) = \{y^*\}^{[r]}(t).$$

Hence special cases of the BPV process include the power variation process and so, obviously, QV.

If $y^* \in \mathcal{SVSM}^c$ and additionally (σ, α^*) are independent of w , then we would expect that for $r+s > 0$,

$$\{y^*\}^{[r,s]}(t) = \mu_r \mu_s \int_0^t \sigma^{r+s}(u) du, \quad \text{where } \mu_r = 2^{r/2} \frac{\Gamma(\frac{1}{2}(r+1))}{\Gamma(\frac{1}{2})}.$$

We will discuss a proof of this general result elsewhere. Again, the importance of the assumed independence assumption is unclear. Overall, this result provides a second way of accessing integrated power volatility, extending previous work on realized power variation. For now we prove a special case.

Theorem 2 Suppose $y^* \in \mathcal{SVSM}^c$ and additionally σ is independent of w and $\alpha^* = 0$. Then, for any $r > 0$, we have

$$\{y^*\}^{[r,r]}(t) = \mu_r^2 \int_0^t \sigma^{2r}(s) ds. \quad (10)$$

Proof It is convenient to introduce the notation

$$\{y_\delta^*\}^{[r,r]}(t) = \delta^{1-r} \sum_{j=1}^{\lfloor t/\delta \rfloor - 1} |y_j|^r |y_{j+1}|^r.$$

Here, and elsewhere, we use δ rather than M as a lower index. The value of the bipower process $\{y^*\}^{[r,r]}$ at time t is the probability limit, as $\delta \downarrow 0$, of $\{y_\delta^*\}^{[r,r]}(t)$. In determining this limit, it causes no loss of generality to assume that t/δ is an integer M , say.

Let $\sigma_j > 0$ be defined by

$$\sigma_j^2 = \sigma^{2*}(j\delta) - \sigma^{2*}((j-1)\delta).$$

We then have

$$\{y_\delta^*\}^{[r,r]}(t) \stackrel{\text{law}}{=} \delta^{1-r} \sum_{j=1}^{M-1} \sigma_j^r \sigma_{j+1}^r |u_j|^r |u_{j+1}|^r,$$

where the u_j are i.i.d. standard normal. This means that

$$\{y_\delta^*\}^{[r,r]}(t) - \mu_r^2 \delta^{1-r} \sum_{j=1}^M \sigma_j^r \sigma_{j+1}^r \stackrel{\text{law}}{=} \delta^{1-r} \sum_{j=1}^{M-1} \sigma_j^r \sigma_{j+1}^r (|u_j| |u_{j+1}| - \mu_r^2).$$

Hence in our proof we need to show three things:

1. As $\delta \downarrow 0$,

$$R = \delta^{1-r} \sum_{j=1}^{M-1} \sigma_j^r \sigma_{j+1}^r (|u_j| |u_{j+1}| - \mu_r^2) \xrightarrow{p} 0. \quad (11)$$

2. As $\delta \downarrow 0$,

$$\delta^{1-r} \left\{ \sum_{j=1}^{M-1} \sigma_j^r \sigma_{j+1}^r - \sum_{j=1}^M \sigma_j^{2r} \right\} \xrightarrow{p} 0. \quad (12)$$

3. As $\delta \downarrow 0$,

$$\delta^{1-r} \sum_{j=1}^M \sigma_j^{2r} \xrightarrow{p} \int_0^t \sigma^{2r}(s) ds.$$

Point 3 is straightforward using Riemann integrability of $\sigma(t)$ [e.g., Barndorff-Nielsen and Shephard (2003d)].

We now take up Part 1. To prove Equation (11), let R' be an independent copy of R and note that $R + R'$ can be rearranged as the sum of two sums each of the form

$$S = \delta^{1-r} \sum_{j=1}^{M-1} \sigma_j^r \sigma_{j+1}^r (w_j - \mu_r^2),$$

where the w_j are i.i.d. and distributed as $|u||u'|$ with u and u' independent standard normal. Barndorff-Nielsen and Shephard (2003c, Corollary 4.3) applies to such a sum, as we indicate below, yielding $S \xrightarrow{p} 0$. Therefore $R + R' \xrightarrow{p} 0$, and as R and R' are independent, the same is true of R .

To see that $S \xrightarrow{p} 0$, let

$$c_{Mj} = \delta^{1-r} \sum_{j=1}^{M-1} \sigma_j^r \sigma_{j+1}^r$$

and

$$x = w - \mu_r^2,$$

where w is standard normal. Then, as is simple to verify, the following three conditions are satisfied:

$$c_M = \max_{1 \leq j < M} c_{Mj} \rightarrow 0$$

$$MP\{|x| \geq c_M^{-1} \varepsilon\} \rightarrow 0$$

$$\sup_M \sum_{j=1}^{M-1} c_{Mj} < \infty,$$

and thus Barndorff-Nielsen and Shephard (2003c, Corollary 4.3) applies.

It only remains to prove Part 2 with Equation (12). In fact, we have the stronger result that

$$\delta^{1-r} \left\{ \sum_{j=1}^{M-1} \sigma_j^r \sigma_{j+1}^r - \sum_{j=1}^M \sigma_j^{2r} \right\} = O(\delta). \quad (13)$$

To establish this, rewrite the difference as follows:

$$\begin{aligned} \sum_{j=1}^{M-1} \sigma_j^r \sigma_{j+1}^r - \sum_{j=1}^M \sigma_j^{2r} &= \sum_{j=1}^{M-1} (\sigma_j^r \sigma_{j+1}^r - \sigma_j^{2r}) - \sigma_M^{2r} \\ &= \sum_{j=1}^{M-1} \sigma_j^r (\sigma_{j+1}^r - \sigma_j^r) - \sigma_M^{2r} \\ &= \sum_{j=1}^{M-1} \frac{\sigma_j^r}{\sigma_{j+1}^r + \sigma_j^r} (\sigma_{j+1}^{2r} - \sigma_j^{2r}) - \sigma_M^{2r} \\ &= \frac{1}{2} \sum_{j=1}^{M-1} \left(\frac{2\sigma_j^r}{\sigma_{j+1}^r + \sigma_j^r} - 1 \right) (\sigma_{j+1}^{2r} - \sigma_j^{2r}) - \frac{1}{2} (\sigma_1^{2r} + \sigma_M^{2r}) \\ &= -\frac{1}{2} \sum_{j=1}^{M-1} \frac{\sigma_{j+1}^r - \sigma_j^r}{\sigma_{j+1}^r + \sigma_j^r} (\sigma_{j+1}^{2r} - \sigma_j^{2r}) - \frac{1}{2} (\sigma_1^{2r} + \sigma_M^{2r}) \\ &= -\frac{1}{2} \sum_{j=1}^{M-1} \frac{(\sigma_{j+1}^{2r} - \sigma_j^{2r})^2}{(\sigma_{j+1}^r + \sigma_j^r)^2} - \frac{1}{2} (\sigma_1^{2r} + \sigma_M^{2r}). \end{aligned}$$

Further, letting $\psi_j = \delta^{-1/2}\sigma_j$, we find

$$\delta^{1-r} \left\{ \sum_{j=1}^{M-1} \sigma_j^r \sigma_{j+1}^r - \sum_{j=1}^M \sigma_j^{2r} \right\} = -\frac{1}{2} \delta \left\{ \sum_{j=1}^{M-1} \frac{(\psi_{j+1}^{2r} - \psi_j^{2r})^2}{(\psi_{j+1}^r + \psi_j^r)^2} + \psi_1^{2r} + \psi_M^{2r} \right\}. \quad (14)$$

By Assumption 1 we have, in particular, that

$$0 < \inf_{0 \leq s \leq t} \sigma(s) \leq \inf \psi_j \leq \sup \psi_j \leq \sup_{0 \leq s \leq t} \sigma(s) < \infty$$

uniformly in δ . From this it is immediate that $\psi_1^{2r} + \psi_M^{2r}$ is uniformly bounded from above. Furthermore,

$$\sum_{j=1}^{M-1} \frac{(\psi_{j+1}^{2r} - \psi_j^{2r})^2}{(\psi_{j+1}^r + \psi_j^r)^2} \leq \frac{1}{4 \inf_{0 \leq s \leq t} \sigma^{2r}(s)} \sum_{j=1}^{M-1} (\psi_{j+1}^{2r} - \psi_j^{2r})^2,$$

where $\inf_{0 \leq s \leq t} \sigma^{2r}(s) > 0$ and, by the càdlàg property of σ ,

$$\sum_{j=1}^{M-1} (\psi_{j+1}^{2r} - \psi_j^{2r})^2 \xrightarrow{P} [\sigma^{2r}](t),$$

which, together with Equation (14), implies Equation (13) and hence Equation (10). ■

Following the initial draft of this article, Barndorff-Nielsen and Shephard (2003a) have shown that the result is not effected by replacing the $\alpha^* = 0$ condition by a much weaker smoothness assumption on α^* .

The above result implies the realized bipower variation

$$\mu_r^{-1} \mu_s^{-1} \{y_M^*\}_i^{[r,s]} \xrightarrow{P} \int_{h(i-1)}^{hi} \sigma^{r+s}(u) du.$$

Clearly

$$\mu_r^{-1} \mu_{2-r}^{-1} \{y_M^*\}_i^{[r,2-r]} \xrightarrow{P} \int_{h(i-1)}^{hi} \sigma^2(u) du, \quad r \in (0, 2),$$

is a consistent estimator of integrated variance and so gives the same result as the realized variance. Further, we have the simple result in the $SVSM^c$ case that

$$[y_M^*]_i^{[2]} - \mu_r^{-1} \mu_{2-r}^{-1} \{y_M^*\}_i^{[r,2-r]}(t) \xrightarrow{P} 0, \quad r \in (0, 2).$$

We will see in the next section that when there are jumps, this term will not have a zero probability limit, but rather will converge to a positive, finite quantity. It will become clear that this difference can be used to test for the presence of jumps.

We now introduce an additional regularity condition.

Condition (V') The volatility process σ has the property

$$\lim_{\delta \downarrow 0} \delta^{1/2} \sum_{j=1}^M |\sigma^r(\eta_j) - \sigma^r(\xi_j)| = 0,$$

for some $r > 0$ (equivalently for every $r > 0$) and for any $\xi_j = \xi_j(\delta)$ and $\eta_j = \eta_j(\delta)$ such that

$$0 \leq \xi_1 \leq \eta_1 \leq \delta \leq \xi_2 \leq \eta_2 \leq 2\delta \leq \dots \leq \xi_M \leq \eta_M \leq t.$$

Using this condition we may strengthen Theorem 2 to the following statement.

Theorem 3 *Let the situation be as in Theorem 2 and suppose moreover that σ satisfies condition (V'). Then*

$$\delta^{1-r} \sum_{j=1}^{\lfloor t/\delta \rfloor - 1} \sigma_j^r \sigma_{j+1}^r = \mu_r^2 \int_0^t \sigma^{2r}(s) ds + o_p(\delta^{1/2}). \quad (15)$$

Proof By the proof of Theorem 2, particularly Equation (13), it suffices to note that when Assumption 1 and Condition (V') hold, then together they imply that

$$\delta^{1-r} \sum_{j=1}^M \sigma_j^{2r} - \sigma^{2r*}(t) = o_p(\delta^{1/2}).$$

This latter relation is in fact the content of Barndorff-Nielsen and Shephard (2003c, Lemma 2). ■

The result of Equation (15) given in Theorem 3 is essential in deriving distributional limit results for realized power variation. See Section 5.4 below. Barndorff-Nielsen and Shephard (2003a) have recently shown that in the case where $r = 1$, Condition (V') can be dispensed with.

2 SV PROCESS PLUS LARGE BUT RARE JUMPS

2.1 Rare Jumps and Quadratic Variation

So far we have seen that in $\mathcal{SVSM}^c$ models, realized variance, realized power variation, and realized bipower variation converge to integrals of powers of spot volatility. When we use special cases of realized bipower variation, then the probability limit is the same as that achieved by realized variance. In this section we will see how this result changes when we introduce jumps.

Consider the log-price process

$$y^*(t) = y^{(1)*}(t) + y^{(2)*}(t), \quad (16)$$

with $y^{(1)*} \in \mathcal{SVSM}^c$ and

$$y^{(2)*}(t) = \sum_{i=1}^{N(t)} c_i, \quad (17)$$

where N is a finite-activity, simple counting process⁹ so that $N(t) < \infty$ (for all $t > 0$), and $\{c_i\}$ is a collection of nonzero random variables. In the special case where N is a homogeneous Poisson process and $\{c_i\}$ are i.i.d. from some distribution D , then $y^{(2)*}$ is a compound Poisson process (written $\mathcal{CPP}$). Such jump process models have often been used as components of price processes [e.g., Merton (1976), Andersen, Benzoni, and Lund (2002), Johannes, Polson, and Stroud (2002), and Chernov et al. (2003)].

The QV for this jump plus $\mathcal{SVSM}^c$ process is well known [see, e.g., the discussion in Andersen, et al. (2001)] and is reported below in the following proposition.

Proposition 1 Suppose $y^{(1)*} \in \mathcal{SVSM}^c$ and $y^{(2)*}$ is given in Equation (17), then

$$\begin{aligned} [y^*](t) &= \sigma^{2*}(t) + \sum_{i=1}^{N(t)} c_i^2 \\ &= [y^{(1)*}]^{[2]}(t) + [y^{(2)*}]^{[2]}(t). \end{aligned}$$

Proof Trivial from Equations (6) and (8). ■

2.2 Rare Jumps and Power Variation

We can generalize the above discussion to deal with power variation. This captures the theoretical effect we are concerned with in this article.

Theorem 4 Let $y^* = y^{(1)*} + y^{(2)*}$ with $y^{(1)*}$ and $y^{(2)*}$ being independent. Suppose $y^{(1)*} \in \mathcal{SVSM}^c$, that is, $y^* = \alpha^* + m^*$, and additionally that (σ, α^*) are independent of w and $y^{(2)*}$ is given in Equation (17). Then

$$\mu_r^{-1}\{y^*\}^{[r]}(t) = \begin{cases} \int_0^t \sigma^r(u) du, & r \in (0, 2) \\ [y^*](t), & r = 2 \\ \infty, & r > 2. \end{cases} \quad (18)$$

Remark 2 The same setting has been independently studied recently by Woerner (2002), who reaches the same conclusion in a general semimartingale framework using a different technique. See also Lépingle (1976) for an earlier investigation.

Proof It is clear that as $\delta \downarrow 0$, then generally

$$\sum_{j=1}^{\lfloor t/\delta \rfloor} |y_j^{(2)}(t)|^r \xrightarrow{p} \sum_{i=1}^{N(t)} |c_i|^r,$$

⁹ A finite-activity counting process is a simple counting process that, with probability one, has a finite number of events happening in a fixed interval of time. This rules out infinite activity Lévy processes, such as the normal inverse Gaussian process, which have an infinite number of jumps in finite intervals of time. These processes will be discussed in Section 5.

where

$$y_j^{(2)}(t) = y^{(2)*}(j\delta) - y^{(2)*}((j-1)\delta).$$

It follows from this that when we also normalize the sum, as $\delta \downarrow 0$,

$$\delta^\beta \sum_{j=1}^{\lfloor t/\delta \rfloor} \left| y_j^{(2)}(t) \right|^r \xrightarrow{p} \begin{cases} 0 & \text{for } \beta > 0, \ r \in (0, 2) \\ [y^{(2)*}](t) & \text{for } \beta = 0, \ r = 2 \\ \infty & \text{for } \beta < 0, \ r > 2. \end{cases}$$

Hence the power variation of $y^{(2)*}$ is either zero, the quadratic variation, or infinity, depending upon the value of r . Furthermore,

$$\sum_{j=1}^{\lfloor t/\delta \rfloor} |y_j(t)|^r = \sum_{j=1}^{\lfloor t/\delta \rfloor} |y_j^{(1)}(t)|^r + O_p(N(t)),$$

as there are only $N(t)$ contributions to $y^{(2)*}$. As a result we have that

$$\mu_r^{-1} \delta^\beta \sum_{j=1}^{\lfloor t/\delta \rfloor} |y_j(t)|^r \xrightarrow{p} \begin{cases} \int_0^t \sigma^r(u) \, du, & \text{for } \beta = 1 - r/2, \ r \in (0, 2) \\ [y^*](t), & \text{for } \beta = 0, \ r = 2 \\ \infty, & \text{for } \beta < 0, \ r > 2. \end{cases}$$

This delivers the required result by definition of $\{y^*\}^{[r]}(t)$. ■

Theorem 4 implies that:

1. When $r \in (0, 2)$, the probability limit of realized power variation is unaffected by the presence of the jumps.
2. When $r > 2$, the probability limit is determined by the jump component and so scaling means this limit is infinite.
3. When $r = 2$, both components contribute.

The result is not very sensitive, for it would hold for any jump process as long as there are a finite number of jumps in any finite period of time. In particular, then, the jumps can be serially dependent, or dependent on the filtration generated by w and σ . The above implies that when $r < 2$ and σ is continuous,¹⁰ then $\sigma(t) = (\mu_r^{-1}(\partial\{y^*\}^{[r]}(t)/\partial t))^{1/r}$.

2.3 Rare Jumps and Bipower Variation

The logic of the proof of Theorem 4 carries over to the bipower variation process. In particular, we deliver the stimulating result on bipower given in the following theorem.¹¹ The only part of the result we will use in practice is the $\max(r, s) < 2$ case.

¹⁰ Otherwise when $r < 2$, then $(\mu_r^{-1} D_- \{y^*\}^{[r]}(t))^{1/r}$ is $\sigma(t-)$.

¹¹ We thank Xin Huang from Duke University for pointing out an error in an earlier version of this result and for deriving the $\max(r, s) = 2$ case.

Theorem 5 *Let the setting be as in Theorem 4. Then, for $r > 0$ and $s > 0$,*

$$\mu_r^{-1} \mu_s^{-1} \{y^*\}^{[r,s]}(t) = \begin{cases} \int_0^t \sigma^{r+s}(u) \, du, & \max(r, s) < 2, \\ x^*(t), & \max(r, s) = 2, \\ \infty, & \max(r, s) > 2, \end{cases} \quad (19)$$

where $x^*(t)$ is some stochastic process.

Proof Defining

$$d_{j,j+1} = \begin{cases} 1, & y_j^{(2)} = y_{j+1}^{(2)} = 0 \\ 1, & y_j^{(2)} \neq 0, y_{j+1}^{(2)} \neq 0, \\ 0, & \text{elsewhere} \end{cases}$$

we have an indicator that is one either if there are no jumps or the jumps are contiguous. Then

$$\begin{aligned} & \delta^{1-(r+s)/2} \sum_{j=1}^{\lfloor t/\delta \rfloor - 1} |y_j|^r |y_{j+1}|^s \\ &= \delta^{1-(r+s)/2} \sum_{j=1}^{\lfloor t/\delta \rfloor - 1} |y_j|^r |y_{j+1}|^s d_{j,j+1} + \delta^{1-(r+s)/2} \sum_{j=1}^{\lfloor t/\delta \rfloor - 1} |y_j|^r |y_{j+1}|^s (1 - d_{j,j+1}). \end{aligned} \quad (20)$$

The impact of the jumps on the first term in Equation (20) of any contiguous jumps goes to zero as $\delta \downarrow 0$ so long as $r + s < 2$, for the jumps contribute a finite number of terms to the sum and so

$$\delta^{1-(r+s)/2} \sum_{j=1}^{\lfloor t/\delta \rfloor - 1} |y_j|^r |y_{j+1}|^s d_{j,j+1} - \delta^{1-(r+s)/2} \sum_{j=1}^{\lfloor t/\delta \rfloor - 1} |y_j^{(1)}|^r |y_{j+1}^{(1)}|^s \xrightarrow{p} 0,$$

as the contribution from the contiguous jumps is killed by the scaling $\delta^{1-(r+s)/2} \downarrow 0$ as $\delta \downarrow 0$. The condition on r and s can be relaxed to $\max(r, s) < 2$ as long as the probability of two contiguous jumps goes to zero at rate δ , which is indeed true for simple counting processes. Of course, under the conditions of the theorem,

$$\delta^{1-(r+s)/2} \sum_{j=1}^{\lfloor t/\delta \rfloor - 1} |y_j^{(1)}|^r |y_{j+1}^{(1)}|^s \xrightarrow{p} \int_0^t \sigma^{r+s}(u) \, du$$

for all r and s .

We now work with

$$\sum_{j=1}^{\lfloor t/\delta \rfloor - 1} |y_j|^r |y_{j+1}|^s (1 - d_{j,j+1}) = O(N(t)),$$

as the terms in the sum are zero unless there is a jump and there are $N(t)$ jumps. The focus of attention is on

$$\begin{aligned}
 & \delta^{1-(r+s)/2} \sum_{j=1}^{\lfloor t/\delta \rfloor - 1} |y_j|^r |y_{j+1}|^s (1 - d_{j,j+1}) \\
 & \leq \delta^{1-(r+s)/2} \sum_{j=1}^{\lfloor t/\delta \rfloor - 1} \left| y_j^{(1)} \right|^r \left| y_{j+1}^{(1)} + y_{j+1}^{(2)} \right|^s (1 - d_{j,j+1}) \\
 & \quad + \delta^{1-(r+s)/2} \sum_{j=1}^{\lfloor t/\delta \rfloor - 1} \left| y_j^{(1)} + y_j^{(2)} \right|^r \left| y_{j+1}^{(1)} \right|^s (1 - d_{j,j+1}) \\
 & = \delta^{1-s/2} \sum_{j=1}^{\lfloor t/\delta \rfloor - 1} \left| \frac{y_j^{(1)}}{\sqrt{\delta}} \right|^r \left| y_{j+1}^{(1)} + y_{j+1}^{(2)} \right|^s (1 - d_{j,j+1}) \\
 & \quad + \delta^{1-r/2} \sum_{j=1}^{\lfloor t/\delta \rfloor - 1} \left| y_j^{(1)} + y_j^{(2)} \right|^r \left| \frac{y_{j+1}^{(1)}}{\sqrt{\delta}} \right|^s (1 - d_{j,j+1}).
 \end{aligned}$$

So long as $s < 2$, then

$$\delta^{1-s/2} \sum_{j=1}^{\lfloor t/\delta \rfloor - 1} \left| \frac{y_j^{(1)}}{\sqrt{\delta}} \right|^r \left| y_{j+1}^{(1)} + y_{j+1}^{(2)} \right|^s (1 - d_{j,j+1}) \xrightarrow{p} 0,$$

while likewise if $r < 2$, then

$$\delta^{1-r/2} \sum_{j=1}^{\lfloor t/\delta \rfloor - 1} \left| y_j^{(1)} + y_j^{(2)} \right|^r \left| \frac{y_{j+1}^{(1)}}{\sqrt{\delta}} \right|^s (1 - d_{j,j+1}) \xrightarrow{p} 0.$$

This delivers the first result in Equation (19), while if $\max(r, s) > 2$, both of these terms explode, which is clearly enough to guarantee the third result in Equation (19).

The only issue that remains is when $\max(r, s) = 2$. In that case

$$\delta^{1-(r+s)/2} \sum_{j=1}^{\lfloor t/\delta \rfloor - 1} |y_j|^r |y_{j+1}|^s (1 - d_{j,j+1})$$

has a stochastic limit that combines the jumps and the volatility process. ■

Remark 3 This result has a special case, of a great deal of applied interest. When $r \in (0, 2)$, then

$$\mu_r^{-1} \mu_{2-r}^{-1} \{y^*\}^{[r, 2-r]}(t) = \int_0^t \sigma^2(u) du,$$

the integrated variance. Hence the integrated variance can be consistently estimated in the presence of rare but large jumps. It also immediately implies that

$$[y^*]^{[2]}(t) - \mu_r^{-1} \mu_{2-r}^{-1} \{y^*\}^{[r, 2-r]}(t) = \sum_{i=1}^{N(t)} c_i^2,$$

the jump contribution to the QV. We believe this is the first analysis that has been able to feasibly decompose the quadratic variation into the contributions from the continuous and jump components of the log-price. Trivially we also have that

$$\mu_r^{-1} \mu_{2-r}^{-1} \{y^*\}^{[r, 2-r]}(t) = \int_0^t \text{Var}(dm^{*c}(u) | \mathcal{F}(u)).$$

Remark 4 The above results imply that realized bipower variation has the feature that when $r \in (0, 2)$, then

$$\mu_r^{-1} \mu_{2-r}^{-1} \{y_M^*\}^{[r, 2-r]} \xrightarrow{p} \int_{h(i-1)}^{hi} \sigma^2(u) du = \int_{h(i-1)}^{hi} \text{var}(dm^{*c}(u) | \mathcal{F}(u)), \quad (21)$$

while

$$\{y_M^*\}_i^{[2]} - \mu_r^{-1} \mu_{2-r}^{-1} \{y_M^*\}_i^{[r, 2-r]} \xrightarrow{p} \sum_{j=N(h(i-1))+1}^{N(hi)} c_j^2. \quad (22)$$

Thus the use of these statistics can, in theory, separately identify the risk associated with the diffuse element of prices and the jumps.

3 SOME SIMULATIONS OF SV PLUS LARGE RARE JUMPS

3.1 Basic Simulation

In this section we will illustrate some of these results on realized bipower variation by simulating an SV plus jump process. Throughout we focus on using two special cases of the results given in Equations (21) and (22). In particular we will study

$$\mu_1^{-2} \{y_M^*\}_i^{[1,1]} = \mu_1^{-2} \sum_{j=1}^{M-1} |y_{j,i}| |y_{j+1,i}| \xrightarrow{p} \int_{h(i-1)}^{hi} \sigma^2(u) du,$$

and that

$$[y_M^*]_i^{[2]} - \mu_1^{-2} \{y_M^*\}_i^{[1,1]} = \sum_{j=1}^M y_{j,i}^2 - \mu_1^{-2} \sum_{j=1}^{M-1} |y_{j,i}| |y_{j+1,i}| \xrightarrow{p} \sum_{j=N(h(i-1))+1}^{N(hi)} c_j^2.$$

Our aim is to see if these convergence results are of any relevance in finite samples.

We start with a discussion of the spot volatility process. We use a Feller or Cox, Ingersoll, and Ross (1985) square root process for σ^2 ,

$$d\sigma^2(t) = -\lambda \{\sigma^2(t) - \xi\} dt + \omega \sigma(t) db(\lambda t), \quad \xi \geq \omega^2/2, \quad \lambda > 0,$$

where b is a standard Brownian motion process. The square root process has a marginal distribution,

$$\sigma^2(t) \sim \Gamma(2\omega^{-2}\xi, 2\omega^{-2}) = \Gamma(v, a), \quad v \geq 1,$$

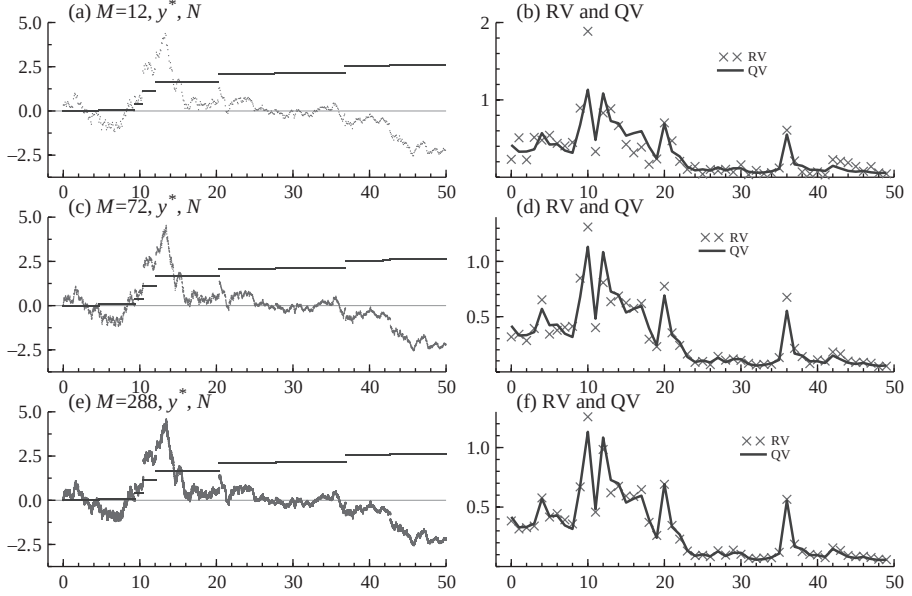


Figure 2 Simulation from a jump plus diffusion-based SV model using a square root variance process. (a), (c), and (e) show $y^*(t)$ and $0.5N(t)$. The latter shows the position of the jumps. (b), (d), and (f) show σ_t^2 and $[y_M^*]_t^{[2]}$.

with a mean of $\xi = \nu/a$, a variance of $\omega^2 = \nu/a^2$, and

$$\text{cor}\{\sigma^2(t), \sigma^2(t+s)\} = \exp(-\lambda|s|).$$

In the context of SV models, this is often called the Heston (1993) model. We take $\alpha^* = 0$ and so

$$y^*(t) = \int_0^t \sigma(u) dw(u) + \sum_{i=1}^{N(t)} c_i.$$

We rule out leverage [e.g., Black (1976) and Nelson (1991)] by assuming $\text{cor}\{b(\lambda t), w(t)\} = 0$. We take $\hbar = 1$, $\lambda = 0.01$, $\nu = 0.1$, and $a = 0.2$. We randomly scatter 10 jumps in a time interval of 50 days, while the jumps are $NID(0, 0.64\nu/a)$. The latter means that when there is a jump, the jump has the same variance as that expected over 64% of a day period of trading when there is no jump. Thus these are large but quite rare jumps.

We report results based on $M = 12$, $M = 72$, and $M = 288$, which are typical practical values for these types of methods. The first row of Figure 2 corresponds to $M = 12$, the second has $M = 72$, and the third $M = 288$. Figure 2a, c, and e shows the sample path of the y^* and N processes recorded at the resolution of M . Notice we have not used a standard time series plot here; rather we plot the processes using dots so that it is possible to see the jumps. As M increases, the sample path of the discretized process better reveals the jump in the process.

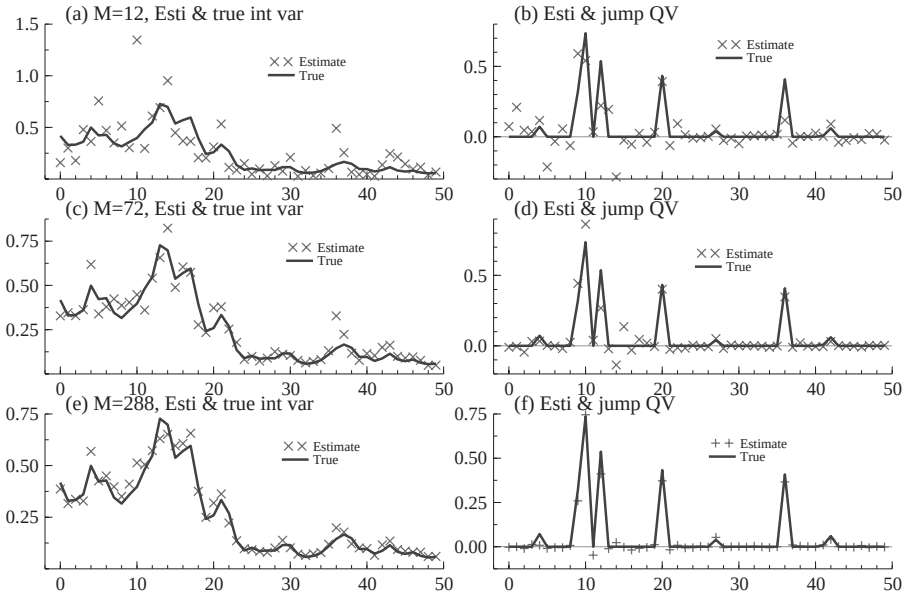


Figure 3 Simulation from a jump plus diffusion-based SV model, estimating the integrated variance and the QV of the jump components. Rows correspond to $M = 12$, $M = 72$, and $M = 288$, respectively. (a), (c), and (e) plot the daily integrated variance σ_t^2 and the consistent estimator $\mu_1^{-2} \{y_M^*\}_i^{[1,1]}$. (b), (d), and (f) plot the QV of the jump process and the corresponding consistent estimator.

Figure 2b, d, and f shows the sample paths of the daily realized variances, while we also draw the corresponding daily increments in the quadratic variation, which is obviously the sum of the integrated variance and the sums of squared jumps. We see that the RV becomes more accurate with M , as we expect from theory. However, we can also see that it is quite an inaccurate estimator.

Figure 3a, c, and e shows the more innovative results. It displays our estimator of the integrated variance, using the theory of realized bipower variation. In particular, as mentioned above, we are using $r = s = 1$ in this work. This is contrasted with the actual increment of the integrated variance. We can see that when $M = 12$, this is a poor estimator and is, in particular, influenced by the large jumps that appear in the sample. However, for moderate M , the statistic seems to be quite informative, while when $M = 288$ the estimator is reasonably accurate. Figure 3b, d, and f shows the difference between the realized variance and the realized bipower variation. The theory suggests this is a consistent estimator of the quadratic variation of the jump component. For small M it is very inaccurate, but for moderate and large M the estimator works rather well.

We back up these results by repeating the above analysis, but using 5000 days, having jumps of the same size as before, but now on average every 20 days. We then focus on the distribution of the estimation errors. In particular we look in

Table 1 Finite sample behavior of the estimators of the quadratic variation of log prices, integrated variance, and quadratic variation of the jump process.

	$[y^*]_i - [y_M^*]_i^{[2]}$			$[y^{(1)*}]_i - \mu_1^{-2} \{y_M^*\}_i^{[1,1]}$			$[y^{(2)*}]_i - [\widehat{y_M^{(2)*}}]_i^{[2]}$		
	Mean	0.025	0.975	Mean	0.025	0.975	Mean	0.025	0.975
M = 12	.001	-.294	.262	-.000	-.338	.299	.002	-.179	.168
M = 48	.001	-.192	.204	-.001	-.239	.207	.003	-.097	.129
M = 72	.001	-.107	.117	-.000	-.134	.130	.001	-.067	.075
M = 144	.000	-.076	.079	-.001	-.098	.093	.001	-.043	.057
M = 288	.000	-.056	.059	-.000	-.066	.064	.000	-.033	.037
M = 576	-.000	-.043	.041	-.001	-.048	.043	.000	-.020	.027

Reported are the errors. We calculate the mean value and the 0.025 and 0.975 quantiles of the sampling distribution.

detail at

$$[y^*]_i - [y_M^*]_i^{[2]},$$

the realized variance error,

$$[y^{(1)*}]_i - \mu_1^{-2} \{y_M^*\}_i^{[1,1]} = \sigma_i^2 - \mu_1^{-2} \sum_{j=1}^{M-1} |y_{j,i}| |y_{j+1,i}|,$$

the realized bipower variation error, and defining

$$[\widehat{y_M^{(2)*}}]_i^{[2]} = \sum_{j=1}^M y_{j,i}^2 - \mu_1^{-2} \sum_{j=1}^{M-1} |y_{j,i}| |y_{j+1,i}|,$$

we also examine

$$[y^{(2)*}]_i - [\widehat{y_M^{(2)*}}]_i^{[2]} = ([y^*]_i - \sigma_i^2) - \left(\sum_{j=1}^M y_{j,i}^2 - \mu_1^{-2} \sum_{j=1}^{M-1} |y_{j,i}| |y_{j+1,i}| \right),$$

the realized jump error. The results are given in Table 1. This records the mean error to see if the estimators are unbiased, as well as the 0.025 and 0.975 quantiles.

For each measure there is little bias, even for small M . In the realized variance case the 95% range of the error falls quite quickly, although the interval is still substantial when $M=576$. An interesting feature is that the errors are roughly symmetrical in this analysis. For the realized bipower variation error the results are rather similar to the RV case. Interestingly, the 95% interval is larger for this error than for the corresponding RV case. For the realized jump error, the 95% intervals are roughly symmetrical.

Table 2 Finite sample behavior of the estimators of the quadratic variation of log prices and integrated variance in the case where there are no jumps.

	$[y^*]_i - [y_M^*]_i^{[2]}$			$\sigma_i^2 - \mu_1^{-2}\{y_M^*\}_i^{[1,1]}$		
	Mean	0.025	0.975	Mean	0.025	0.975
M = 12	.001	-.284	.258	.002	-.324	.299
M = 48	.001	-.189	.201	.000	-.220	.209
M = 72	.001	-.106	.113	.001	-.130	.130
M = 144	.000	-.074	.079	.000	-.089	.093
M = 288	-.000	-.055	.056	.000	-.061	.064
M = 576	-.000	-.042	.040	-.000	-.046	.043

Reported are the errors. We calculate the mean value and the 0.025 and 0.975 quantiles of the sampling distribution.

3.2 No Jump Case

Interestingly, when we repeat the analysis, removing all jumps and comparing the performance of $[y_M^*]_i^{[2]}$ and $\mu_1^{-2}\{y_M^*\}_i^{[1,1]}$ as estimators of σ_i^2 , we obtained the results given in Table 2 that indicate the RV is slightly preferable in terms of accuracy. A theoretical confirmation of this result is indicated in Section 5 of this article.

3.3 Improving the Finite Sample Behavior

It is clear that

$$[y^{(2)*}]_i = [y^*]_i - \sigma_i^2 = \sum_{j=N\{h(i-1)\}+1}^{N(hi)} c_j^2 \geq 0,$$

while the estimator

$$[\widehat{y_M^{(2)*}}]_i^{[2]} = [y_M^*]_i^{[2]} - \mu_1^{-2}\{y_M^*\}_i^{[1,1]}$$

can be negative. Hence it would be sensible to sometimes replace it by the consistent estimator

$$\max\left([y_M^*]_i^{[2]} - \mu_1^{-2}\{y_M^*\}_i^{[1,1]}, 0\right). \quad (23)$$

This would suggest the following estimator of the actual variance of the SV component

$$\min\left\{[y_M^*]_i^{[2]}, \mu_1^{-2}\{y_M^*\}_i^{[1,1]}\right\}.$$

Repeating the above experiment based on 5000 days for the nonlinear estimator of the QV of the jump component of Equation (23), we produce the analysis of errors given in Table 3. We see that the finite sample behavior of the estimator improves quite considerably. Importantly, the latter term now has a very short right-hand tail, which means we rarely overstate the presence of jumps by a large amount. Of course, this estimator has the disadvantage of being biased.

Table 3 Improved finite sample behavior of the estimator of the quadratic variation of the jump process.

	Jump error		
	Mean	0.025	0.975
M = 12	-.016	-.179	.025
M = 48	-.010	-.097	.017
M = 72	-.006	-.067	.011
M = 144	-.004	-.043	.011
M = 288	-.003	-.033	.008
M = 576	-.001	-.020	.006

Reported are the errors. We calculate the mean value and the 2.5% and 97.5% quantiles of the sampling distribution.

4 INITIAL EMPIRICAL WORK

To illustrate some of the new theory, we return to the dataset discussed in the introduction. We now work with the full dataset from December 1, 1986, until November 30, 1996. All calculations will be based on $M = 288$, that is we will employ five-minute returns, calculating aggregate volatility statistics for each day within the sample.

Figure 4a shows the time series of the daily realized variance, while Figure 4b shows the corresponding realized bipower variation. The difference between these two estimates, which is our estimate of the daily increments of QV of the jump component, is given in Figure 4c. Finally, Figure 4d shows the correlograms of the realized variances, the realized bipower variations, and the estimated jump components of QV. The main feature of Figure 4 is that the estimate of the integrated variance is the dominating component of the realized variance. Some of the estimates of the jump QV are quite large, especially at the start of the period. However, these jumps tend not to be very serially dependent. The serial dependence in the estimated integrated variances is larger than for the RVs. This is interesting, although the increase in dependence is quite modest. During this period the average value of the jump component to daily QV is 0.0310. Average realized variance is 0.528, hence the size of the jump component is quite modest for the dataset.

Figure 5 focuses on two time periods with large realized variances.¹² In both cases we look at 10-day periods of time, plotting in Figure 5a and c the log-price process every five minutes. Throughout Figure 5a and c we plot the prices using

¹² The result does not change whether one uses the Andersen et al. (2001) version of this data or the dataset that includes the Barndorff-Nielsen and Shephard (2002b) stochastic adjustments.

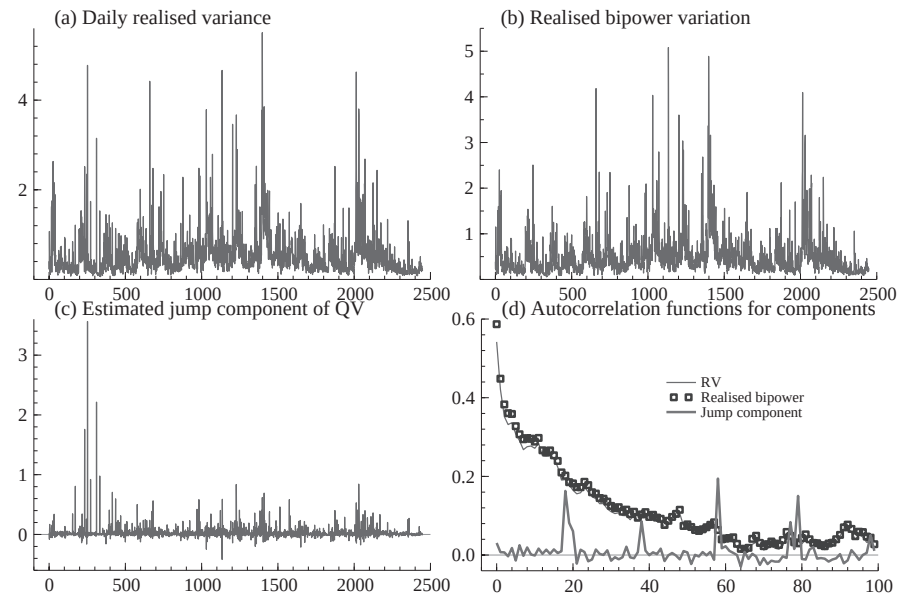


Figure 4 DM/dollar daily volatility measurements based on $M = 288$.

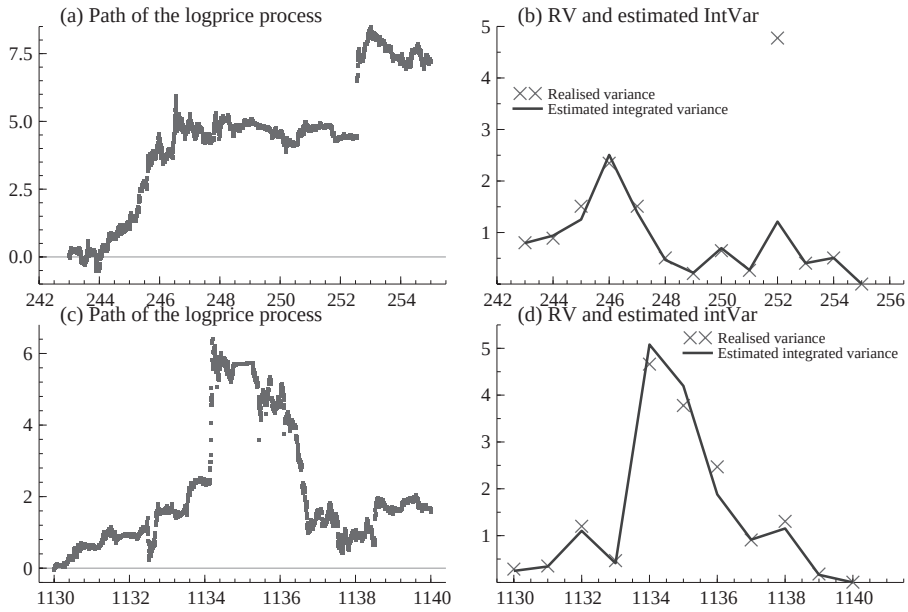


Figure 5 Two examples of small stretches of data with large realized variances. The log-prices are recorded every five minutes in (a) and (c). (a) has a large step change in the prices, which one may view as a jump. The change occurred on January 15, 1988, when surprising U.S. balance of payments were announced. (c) has a number of large positive returns that are in sequence. This happened on August 19, 1991, the day of the coup against Gorbachev.

dots rather than the more standard time-series lines. In Figure 5a there is a large uptick in the price, with a movement of nearly 2% in a five-minute period. This occurred on January 15, 1988. The *Financial Times* reported on its front page the next day, "The dollar and share prices soared in hectic trading on world financial markets yesterday after the release of official figures showing that the US trade deficit had fallen to \$13.22 bn in November from October's record level of \$17.63 bn. The US currency surged 4 pfennigs and 4 yen within 10 minutes of the release of the figures and maintained the day's highest levels in late New York business . . ." The 15th had a large realized variance, shown in Figure 5b, but a much smaller estimate of the integrated variance. Hence the statistics are attributing a large component of the realized variance to the jump.

This effect contrasts with Figure 5c, where there is a three unit increase in the log-price, but this happens over about an hour long-period with many positive returns. This happened on August 19, 1991. In the early hours of that day Russian President Mikhail Gorbachev was overthrown after a coup by Communist hard-liners. We can see in Figure 5d that the corresponding realized variance is very high, but so is the estimated integrated variance. Thus in this case the statistics have not flagged up a jump in the price, even though prices were moving rapidly in one direction.

5 EXTENSIONS AND DISCUSSION

5.1 Robust Estimation of Integrated Covariance

A vital concept in financial econometrics is covariance. We can formally base this on the concept of quadratic covariation. We will discuss this in the context of a bivariate semimartingale (x^*, y^*) . Then the quadratic covariation between x^* and y^* is

$$[y^*, x^*](t) = p - \lim_{M \rightarrow \infty} \sum_{j=1}^M \{y^*(t_j) - y^*(t_{j-1})\} \{x^*(t_j) - x^*(t_{j-1})\}.$$

Note that using this notation

$$[y^*, y^*](t) = [y^*](t),$$

the QV of the y^* process. Quadratic covariation has been recently studied in the econometric literature by, for example, Barndorff-Nielsen and Shephard (2002a) and Andersen et al. (2003).

In the context of multivariate $\mathcal{SVSM}^c$ models [Barndorff-Nielsen and Shephard (2002a)] we assume

$$\begin{pmatrix} y^*(t) \\ x^*(t) \end{pmatrix} = \alpha^*(t) + m^*(t), \quad \text{where } m^*(t) = \int_0^t \Theta(u) dw(u),$$

$\alpha^* \in \mathcal{FV}_{loc}^c$, the elements of Θ are assumed càdlàg, and w is a vector of independent standard Brownian motions. We write $\Sigma(t) = \Theta(t)\Theta(t)'$, and we need to make the additional assumption that $\int_0^t \Sigma_{k,k}(u) du < \infty$, $k=1, 2$, to ensure that m^* is a local martingale. This setup has the important property that

$$[y^*, x^*](t) = \int_0^t \Sigma_{1,2}(u) du,$$

the integrated covariance of the price process. Andersen et al. (2003) emphasize the importance of estimating this type of term and they discuss the use of realized covariation to achieve this. A distribution theory for this estimator under the above type of model structure is given in Barndorff-Nielsen and Shephard (2002a).

An important property of quadratic variation is the so-called polarization result that

$$[y^* + x^*](t) = [y^*](t) + [x^*](t) + 2[y^*, x^*](t),$$

which means that

$$[y^*, x^*](t) = \frac{1}{2}\{[y^* + x^*](t) - [y^*](t) - [x^*](t)\}.$$

Hence we can use realized bipower variation to consistently estimate $[y^*, x^*](t)$, or increments of it, by estimating each of the individual terms. Further, straightforwardly, this estimator is robust to jumps and we now have tools for assessing if the dependence between price processes is affected by the presence of jumps. We will explore this issue elsewhere in some detail.

5.2 Generalization to Multipower Variation

We defined the bipower variation as

$$p - \lim_{\delta \downarrow 0} \delta^{1-(r+s)/2} \sum_{j=1}^{\lfloor t/\delta \rfloor - 1} |y_j(t)|^r |y_{j+1}(t)|^s, \quad r, s \geq 0.$$

It is clear we can generalize this object by multiplying together a finite number of absolute returns raised to some nonnegative power. In general we call this idea multipower variation. In particular the tripower variation process is

$$\{y^*\}^{[r,s,u]}(t) = p - \lim_{\delta \downarrow 0} \delta^{1-(r+s+u)/2} \sum_{j=1}^{\lfloor t/\delta \rfloor - 2} |y_j(t)|^r |y_{j+1}(t)|^s |y_{j+2}(t)|^u, \quad r, s, u > 0,$$

while the quadpower variation process is

$$\{y^*\}^{[r,s,u,v]}(t) = p - \lim_{\delta \downarrow 0} \delta^{1-(r+s+u+v)/2} \sum_{j=1}^{\lfloor t/\delta \rfloor - 3} |y_j(t)|^r |y_{j+1}(t)|^s |y_{j+2}(t)|^u |y_{j+3}(t)|^v, \\ r, s, u, v > 0.$$

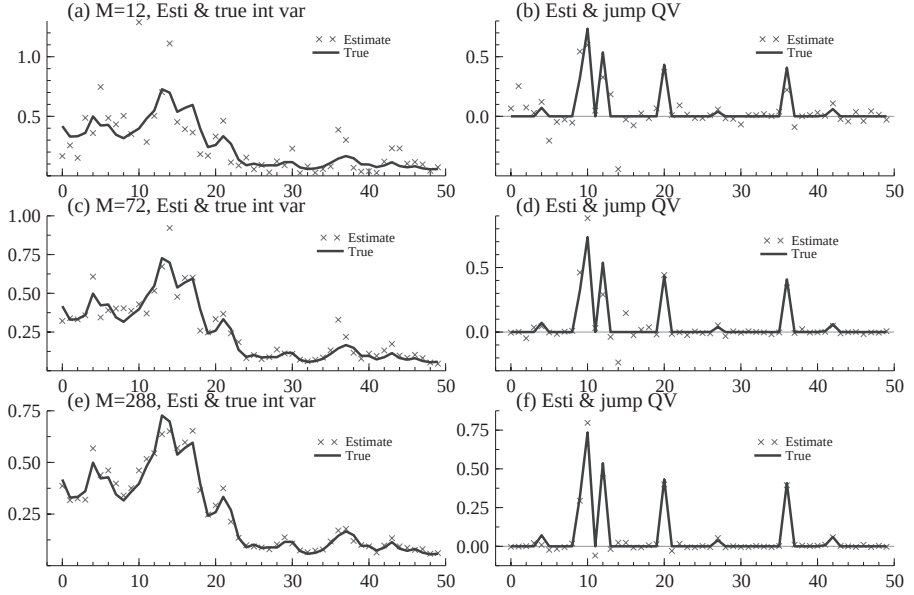


Figure 6 Estimation based on tripower variation. Simulation from a jump plus diffusion-based SV model. Rows correspond to $M = 12$, $M = 72$, and $M = 288$, respectively. (a), (c), and (e) plot the daily integrated variance σ_t^2 and the consistent estimator $\mu_{2/3}^{-3}\{y_M^*\}^{[2/3, 2/3, 2/3]}$. (b), (d), and (f) plot the QV of the jump component and the corresponding consistent estimator.

Example 1 Taking $r = s = u = 2/3$ yields

$$\mu_{2/3}^{-3}\{y^*\}^{[2/3, 2/3, 2/3]}(t) = \int_0^t \sigma^2(u) du,$$

another new estimator of integrated variance. Figure 6 repeats the experiments reported in Figure 3, but replaces bipower with tripower. In particular we have plotted the time series of the realized tripower term

$$\mu_{2/3}^{-3} \sum_{j=1}^{M-2} |y_{j,i}|^{2/3} |y_{j+1,i}|^{2/3} |y_{j+2,i}|^{2/3},$$

instead of

$$\mu_1^{-2} \sum_{j=1}^{M-1} |y_{j,i}| |y_{j+1,i}|.$$

Likewise

$$\mu_{1/2}^{-4}\{y^*\}^{[1/2, 1/2, 1/2, 1/2]}(t) = \int_0^t \sigma^2(u) du.$$

This result is interesting, as it estimates integrated variance based on square roots of absolute returns, which is quite a robust item. Figure 7 again repeats the

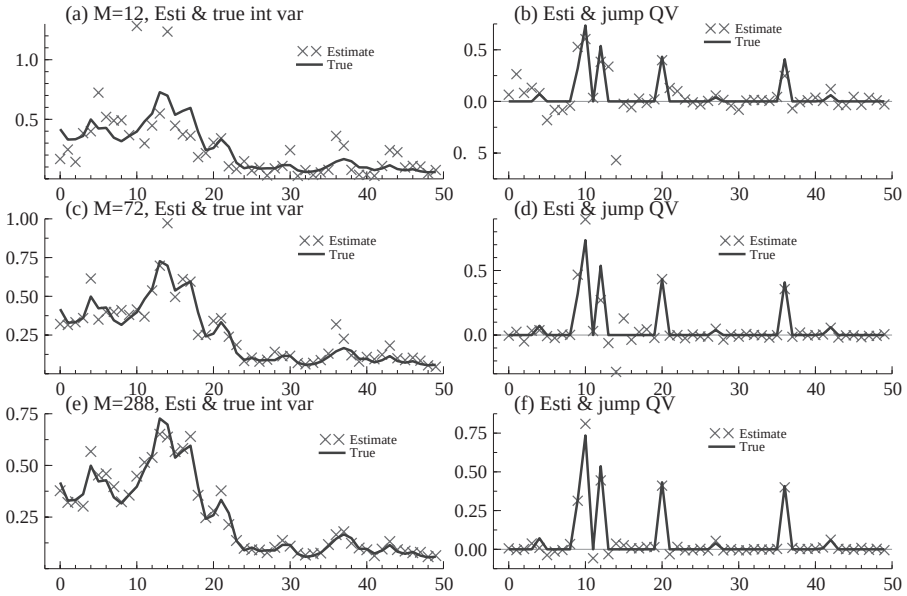


Figure 7 Estimation based on quadpower variation. Simulation from a jump plus diffusion-based SV model. Rows correspond to $M=12$, $M=72$, and $M=288$, respectively. (a), (c), and (e) plot the daily integrated variance σ_t^2 and the consistent estimator $\mu_{1/2}^{-4} \{y_M^*\}_t^{[1/2, 1/2, 1/2, 1/2]}$. (b), (d), and (f) plot the QV of the jump component and the corresponding consistent estimator.

previous experiments, but now based on this quadpower concept. The realized version of the quadpower is

$$\mu_{1/2}^{-4} \sum_{j=1}^{M-3} \sqrt{|y_{j,i} y_{j+1,i} y_{j+2,i} y_{j+3,i}|}.$$

We see from Figure 7 that we do not produce very different answers. An important feature of this graph is that for one period the estimate of the QV of the jump component is highly negative. This makes little sense, as we know the jump component has to be nonnegative. We discussed this issue in the context of bipower in Section 3.3. If we use the same strategy here, then we produce much more sensible results. They are plotted in Figure 8. This suggests that the estimators are reasonably accurate even for quite small samples.

Example 2 Taking $r=s=2$ yields

$$\{y^*\}^{[2,2]}(t) = \int_0^t \sigma^4(u) du,$$

the integrated quarticity. The realized version of this was used by Barndorff-Nielsen and Shephard (2002a) to construct an asymptotic

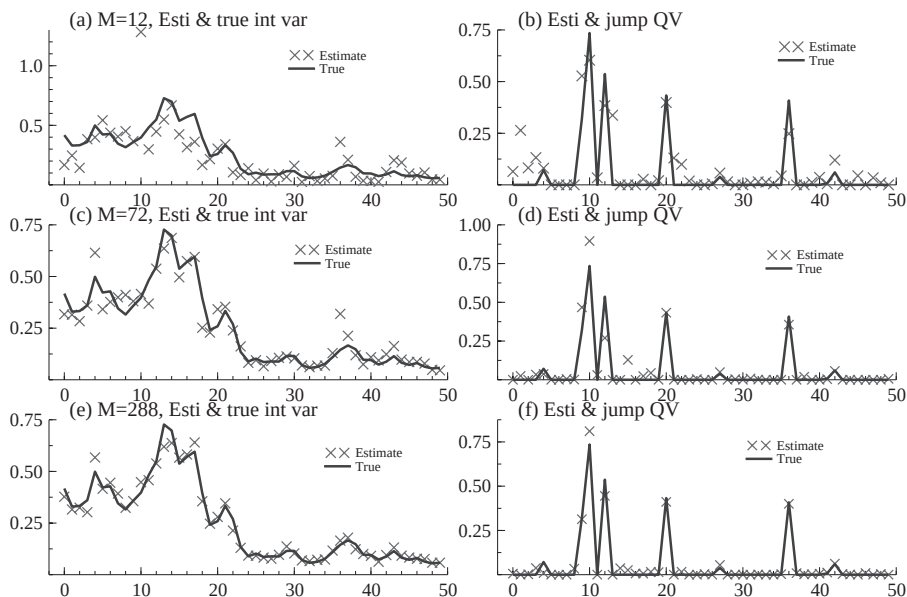


Figure 8 Estimation based on the truncated version of quadpower variation, which is enforced to improve the finite sample behavior of the estimator. Simulation from a jump plus diffusion based SV model. Rows correspond to $M = 12$, $M = 72$, and $M = 288$, respectively. (a), (c), and (e) plot the daily integrated variance σ_t^2 and the consistent estimator. (b), (d), and (f) plot the jump QV and consistent estimator.

distribution theory for realized variance and covariation. A more robust object of this type is

$$\mu_1^{-4}\{y^*\}^{[1,1,1,1]}(t) = \int_0^t \sigma^4(u) du.$$

We will return to this in a moment when we discuss testing for jumps.

5.3 Some Infinite Activity Lévy Processes

In our analysis we have shown that the realized power variation of a compound Poisson process converges in probability to zero when the power is less than two. An interesting extension of this result is to Lévy processes with infinite numbers of jumps in finite time intervals. Then the question is whether bipower variation is robust to these types of jumps. Here we explore this problem, without giving a complete solution.

Recall Lévy processes are processes with independent and stationary increments. Examples of this type of process are the normal gamma (or variance gamma) process and the normal inverse Gaussian process. These are due to Madan and Seneta (1990) and Madan, Carr, and Chang (1998) in the former case and Barndorff-Nielsen (1997) and Barndorff-Nielsen (1998) in the latter case.

Let z denote a Lévy process. It is characterized by the Lévy-Khintchine representation, which says that we can write

$$\log E \exp\{i\zeta z(1)\} = ai\zeta - \frac{1}{2}\sigma^2\zeta^2 - \int_{\mathbf{R}} \{1 - e^{i\zeta x} + i\zeta x \mathbf{1}_B(x)\} W(dx), \quad (24)$$

where $a \in \mathbf{R}$, $\sigma \geq 0$, $B = [-1, 1]$ and the Lévy measure W must satisfy

$$\int_{\mathbf{R}} \min\{1, x^2\} W(dx) < \infty \quad (25)$$

and W has no atom at zero.

In this exposition we will assume the drift a and the volatility σ are zero, so z is a pure jump process. The Lévy measure controls the jumps in the process. If $\int_{\mathbf{R}} W(dx) < \infty$, then the process is a compound Poisson process and so has a finite number of jumps. If this does not hold but $\int_{\mathbf{R}} \min\{1, |x|\} W(dx) < \infty$, then we say the process is of infinite-activity finite variation (IAFV). It has an infinite number of jumps in any finite time interval. However, the finite variation feature of the process means that it can be decomposed into

$$z(t) = z_+(t) - z_-(t),$$

where z_+ and z_- are independent subordinators. Recall a subordinator is a Lévy process with nonnegative increments. Under the condition Equation (25), this decomposition into positive and negative jumps is not possible in general.

We believe that the results of Section 2 extend to cases where the jump process is of the Lévy IAFV type, and a discussion of this will be provided elsewhere. Following the initial draft of this article, Woerner (2003) formally studied robustness results of this type, extending the derivations we have given to cases that allow infinite numbers of jumps in finite time periods.

5.4 Asymptotic Distribution

An important question is whether we can extend the convergence in probability result we saw for realized bipower variation to convergence in distribution. This type of result has been developed for realized variance by Barndorff-Nielsen and Shephard (2002a, 2002b). It has been extended considerably in their work on realized power variation by Barndorff-Nielsen and Shephard (2003c, 2003d). Following the initial draft of this article, Barndorff-Nielsen and Shephard (2003a) have derived the joint asymptotic distribution of realized bipower and quadratic variation in the $r=s=1$ case, under some rather weak assumptions. Here we outline the result, leaving the reader to look at that article for technical conditions. In particular they show that

$$\frac{\delta^{-1/2}}{\sqrt{\int_0^t \sigma^4(u) du}} \begin{pmatrix} \sum_{j=1}^{\lfloor t/\delta \rfloor - 1} y_j^2 - \int_0^t \sigma^2(s) ds \\ \mu_1^{-2} \sum_{j=1}^{\lfloor t/\delta \rfloor - 1} |y_j| |y_{j+1}| - \int_0^t \sigma^2(s) ds \end{pmatrix} \xrightarrow{law} N \left\{ 0, \begin{pmatrix} 2 & 2 \\ 2 & 2.60907 \end{pmatrix} \right\}.$$

An implication is that the correlation between these two measures of integrated variance is constant through time and independent of the model. This analysis suggests that the realized variance statistic is, as expected, slightly more efficient than realized bipower variation. This agrees with the results in Table 2. The correlation between the two statistics is approximately 0.875. The implication is that, under the $\mathcal{SVSM}^c$ model, there is relatively little additional information in the realized bipower variation compared to the realized variance. Indeed, as

$$\begin{pmatrix} 2 & 2 \\ 2 & 2.60907 \end{pmatrix}^{-1} = \begin{pmatrix} 2.1418 & -1.6418 \\ -1.6418 & 1.6418 \end{pmatrix},$$

the generalized least-squares estimator of $\int_0^t \sigma^2(s)ds$, based on the realized variance and realized bipower variation, will put weights on the realized variance and realized bipower variation that are approximately one and zero, respectively.

Straightforwardly we can also look at the behavior of the difference between realized variance and realized bipower variation. The above limit theory immediately implies that

$$\frac{\delta^{-1/2}}{\sqrt{\int_0^t \sigma^4(u)du}} \left(\sum_{j=1}^{\lfloor t/\delta \rfloor - 1} y_j^2 - \mu_1^{-2} \sum_{j=1}^{\lfloor t/\delta \rfloor - 1} |y_j| |y_{j+1}| \right) \xrightarrow{law} N(0, 0.6091).$$

This can be used to test for jumps, using the above as the null distribution. Clearly it may be more desirable to look at the difference of the log of the realized variance and the realized log bipower variation. Using the delta method on the multivariate distribution implies

$$\frac{\delta^{-1/2}}{\sqrt{\left(\int_0^t \sigma^4(u)du \right) / \left\{ \int_0^t \sigma^2(u)du \right\}^2}} \left\{ \log \left(\sum_{j=1}^{\lfloor t/\delta \rfloor - 1} y_j^2 \right) - \log \left(\mu_1^{-2} \sum_{j=1}^{\lfloor t/\delta \rfloor - 1} |y_j| |y_{j+1}| \right) \right\} \xrightarrow{law} N(0, 0.6091).$$

Following the argument in Barndorff-Nielsen and Shephard (2004) we noted that in this type of context it is helpful to use Jensen's inequality, which implies that

$$\frac{\delta \mu_1^{-4} \sum_{j=1}^{\lfloor t/\delta \rfloor - 3} |y_j| |y_{j+1}| |y_{j+2}| |y_{j+3}|}{\left\{ \mu_1^{-2} \sum_{j=1}^{\lfloor t/\delta \rfloor - 1} |y_j| |y_{j+1}| \right\}^2} \xrightarrow{p} \frac{\int_0^t \sigma^4(u)du}{\left\{ \int_0^t \sigma^2(u)du \right\}^2} \geq t^{-1}.$$

This suggests using the asymptotic approximation

$$\frac{\left\{ \log \left(\sum_{j=1}^{\lfloor t/\delta \rfloor - 1} y_j^2 \right) - \log \left(\mu_1^{-2} \sum_{j=1}^{\lfloor t/\delta \rfloor - 1} |y_j| |y_{j+1}| \right) \right\}}{\sqrt{0.6091 \max \left\{ (\delta/t), \sum_{j=1}^{\lfloor t/\delta \rfloor - 3} |y_j| |y_{j+1}| |y_{j+2}| |y_{j+3}| / \left(\sum_{j=1}^{\lfloor t/\delta \rfloor - 1} |y_j| |y_{j+1}| \right)^2 \right\}}} \xrightarrow{law} N(0, 1). \quad (26)$$

Table 4 Olsen data, dollar against the DM.

Proportion of rejections	Critical value	Significance level
0.215	1.64	0.05
0.119	2.32	0.01
0.0600	3.09	0.001
0.0383	3.71	0.0001

Tests for jumps based on asymptotic theory.

To illustrate the above theory we return to the Olsen data discussed above and use a one-sided test to see if there are jumps in the series. Table 4 gives the percentage of rejections for the series, together with the associated critical value and significance levels. The table shows there are about one in five rejections of the hypothesis of no jumps when we use a 5% significance level. This falls to about a 1 in 10 rejection when we move to a 1% level. Following the initial drafts of this article, similar empirical results have been reported by Andersen, Bollerslev, and Diebold (2003), who have also studied the use of the difference between realized variance and bipower variation to test for jumps on this dataset. They have used such tests to then build reduced-form models to improve on existing forecasting devices for predicting future realized variances. Huang and Tauchen (2003) and Barndorff-Nielsen and Shephard (2003a) have independently and concurrently used simulations to study the finite sample performance of these types of tests, reaching broadly similar conclusions that transformed versions of the tests (such as the log version detailed above) are reasonably well behaved in empirically reasonable sample sizes.

5.5 Other Related Work on Jumps

Here we briefly discuss some related econometric work that has studied jump-type models.

There is considerable literature on estimating continuous-time parametric models. A particular focus has been on the class of SV plus jumps model. Johannes, Polson, and Stroud (2002) provide filtering methods for parameterized stochastic differential equations that exhibit finite activity jumps. Their approach can be generalized to deal with infinite activity processes for it relies on the very flexible auxiliary particle filters introduced by Pitt and Shephard (1999), while corresponding filtering work on discrete-time models was carried out by Chib, Nardari, and Shephard (2002). Andersen, Benzoni, and Lund (2002) and Chernov et al. (2003) have used EMM methods to estimate and test some of these models. Maheu and McCurdy (2003) have constructed a jump/GARCH discrete-time model for daily data that attempts to unscramble jumps and volatility changes, while Eraker, Johannes, and Polson (2003) look at the same issue based

on SV models using Markov chain Monte Carlo methods [using the methods of handling diffusions introduced by Eraker (2001) and Elerian, Chib, and Shephard (2001)].

There is also literature on discrete-time parametric models of SV plus jumps. Chib, Nardari, and Shephard (2002) use simulation-based likelihood inference, which allows one to test for jumps in a standard manner. Other articles on this topic are pointed out in that article's references.

Aït-Sahalia (2002) recently asked the generic question of whether a continuous-time Markov process exhibits jumps? He developed a theory for this based on transition densities and then applied this to financial data. In some recent stimulating work Mancini (2001, 2003a, 2003b) has tried to estimate the jump component of quadratic variation. She does this via the introduction of a jump threshold whose absolute value goes to zero as the number of observations within each day goes to infinity.

Finally, and closest to this article, Barndorff-Nielsen and Shephard (2003b) have studied the second-order properties of realized variance under the assumption that the local martingale component of prices is a time-changed Lévy process. Such models will have jumps in the price process except in the Brownian motion special case. For time-changed Lévy processes the realized variance is an inconsistent estimator of the increments of the time change. They used their results to suggest simple quasi-likelihood estimators of these types of models.

6 CONCLUSION

In this article we have studied the recently formalized concept of realized power variation in the context of SV models where there can be occasional jumps. We have shown that sometimes the probability limits of these quantities are unaffected by rare jumps. Realized power variation has inspired us to introduce realized bipower variation. This shares the same robustness property, but can also be set up to estimate integrated power volatility in SV models. In a range of cases this produces an estimator of integrated variance in the presence of jumps. To our knowledge this is the first radical alternative to the commonly used realized variance estimator of this quantity. Importantly, when we add jumps to the SV model the probability limit of the bipower estimator does not change, which means we can combine realized variance with realized bipower variation to estimate the quadratic variation of the jump component.

Various extensions of our work have been developed. We have outlined multivariate versions of the methods, a distribution theory, and an assessment of the robustness to the α^* process. We think our article may open up a number of interesting research avenues for econometricians.

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Discussion

TORBEN G. ANDERSEN

Northwestern University and National Bureau of Economic Research

I am honored and pleased to be given the opportunity to comment on this invited paper by Barndorff-Nielsen and Shephard (hereafter BNS). Beyond commenting on this particular contribution, it gives me an opportunity to reflect on the intuition behind and the origin of the rapidly evolving literature on realized volatility modeling — a development BNS have influenced and helped shape.

1 PERSONAL REFLECTIONS ON THE EVOLUTION OF THE REALIZED VOLATILITY APPROACH

At the risk of appearing self-indulgent, I will try to communicate my own sense of discovery through my involvement with these broad research topics over many years. I feel it is the best way to explain what "all the fuss is about." In the second phase of the process, BNS are a critical ingredient in this — to me — fascinating scientific process. Let me further warn in advance that I make no attempt at citing related work in any comprehensive manner — it is simply not possible in an informal comment of this type.

So I begin by recalling my first empirical experimentations with high-frequency data that inspired Tim Bollerslev and myself to pursue further empirical work on high-frequency volatility measures as well as seeking a theoretical underpinning behind our findings. In the period 1995–1997, we were actively working on reconciling the strong volatility persistence evident in daily financial return data with the seemingly much lower estimates of persistence reported by authors using high-frequency (intraday) data. The latter findings were obtained through standard generalized autoregressive conditionally heteroscedastic (GARCH) or stochastic volatility models applied directly to the intraday data.

Address correspondence to Torben G. Andersen, Nathan S. and Mary P. Sharp Distinguished Professor of Finance, Kellogg School of Management, Northwestern University, 2001 Sheridan Rd., Evanston, IL 60208, or e-mail: t-andersen@kellogg.northwestern.edu.

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